Replicating the eGRID Calibration Step

Fowlie, Petersen & Reguant (2021) — Border Carbon Adjustments When Carbon Intensity Varies Across Producers

Independent replication and follow-on analysis · August 16, 2026

Overview

This is an independent study replicating the paper above, focused on the eGRID2018 thermal-plant calibration step of the paper's simulation pipeline. This step assigns each thermal power plant in the Western Electricity Coordinating Council (WECC) region an emissions intensity and operating characteristics, cleaned from EPA's eGRID2018 database and assigned to one of four model regions (California, Northwest, Southwest, Rockies). It is the calibration input that the paper's full market-equilibrium model builds on to produce Figure 1 and Table 1 in the original paper; a portion with a manageable data size has been replicated.

Methodology

The original authors' **eGrid.py** script (part of their openICPSR replication package, DOI 10.3886/E131024V1) was obtained and run against the public EPA eGRID2018 dataset to produce a cleaned, plant-level thermal generator table. All analysis beyond that point — the five outputs below — is original work: independent Python scripts (pandas/numpy) computing regional summary statistics, a generation-weighted regional average, and a fuel-type decomposition, none of which appear in the original paper's own analysis code.

Findings

1–2. Plant counts and rate distribution by region

577 California plants, 253 Northwest, 283 Southwest, and 189 Rockies (plus 5 unmatched to any of the four model regions). Every region's plant-level emissions-rate distribution is strongly right-skewed — mean far exceeds median in every case — indicating a small number of extreme outliers are dragging up the simple average.

Region	Plants	Mean (lb/MWh)	Median (lb/MWh)	Max (lb/MWh)
California	577	1650.1	905.9	122,001.9
Northwest	253	1568.8	1207.5	12,221.8
Southwest	283	1136.8	1015.3	4,559.7
Rockies	189	1706.9	1176.1	9,121.6

3. Outlier drill-down

The most extreme California rate (122,001.9 lb/MWh) traces to a 24.9 MW gas plant in Stanislaus County that generated only 5.0 MWh across the entire year — a capacity factor near zero. Dividing near-zero annual emissions by an even smaller annual output produces an inflated intensity figure; this reflects a real-world data characteristic of idle peaker plants, not a data-pipeline error.

4. Generation-weighted vs. simple regional average

Weighting each plant's rate by its actual annual generation — rather than treating every plant equally — produces the economically meaningful comparison: what is actually being emitted per MWh really produced, not per plant that happens to exist.

Region	Weighted avg (tonnes CO2/MWh)	Paper reference point
California	0.437	CA default import rate: 0.428
Southwest	0.685	Paper: nonbaseload intensity "above 0.7"
Rockies	0.828	Paper: nonbaseload intensity "above 0.7"
Northwest	0.781	—

California's weighted average lands close to the state's own regulatory default import rate — a coincidental but reassuring magnitude check, since that default was originally calibrated from different (2006–2008, out-of-state producer) data. The Southwest and Rockies figures are consistent with the paper's explicit statement that eGRID nonbaseload carbon intensities exceed 0.7 tonnes/MWh in those two regions.

5. Fuel-type breakdown by region

Explains the mechanism behind finding 4: coal supplies a disproportionately large share of real generation (not merely plant count) in the Northwest and Rockies relative to California, and coal's per-MWh emissions rate is consistently far higher than gas's in every region. This is the concrete data-level illustration of the paper's central motivating claim — that out-of-state producers, particularly in coal-heavier regions, are meaningfully dirtier than California's largely gas-fired fleet, which is exactly the heterogeneity that makes a single flat BCA benchmark rate contentious.

Region	Fuel	Plants	Total gen (MWh)	Mean rate (lb/MWh)
California	GAS	559	88,912,696	1646.6
California	COAL	4	8,793,545	1588.0
Northwest	COAL	43	63,072,487	1997.3
Northwest	GAS	197	45,596,811	1447.7
Southwest	GAS	261	67,001,781	1035.7
Southwest	COAL	18	44,335,727	2340.6
Rockies	COAL	22	29,543,672	2476.8
Rockies	GAS	146	16,338,939	1496.3

Attribution and Data Provenance

The eGRID2018 cleaning script used to produce the underlying plant-level dataset is the original work of Meredith Fowlie, Claire Petersen, and Mar Reguant, deposited as part of their replication package on openICPSR (DOI 10.3886/E131024V1). All analysis presented in this report beyond that cleaning step — the regional distribution statistics, outlier investigation, generation-weighting, and fuel-type decomposition — is independent work.

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